

What's New in the PrimeWave Design Environment Version W-2024.09

The PrimeWave Design Environment W-2024.09 introduces new features, enhancements, and changes in the following product areas.

- [Release Notes and Installation](#)
- [PrimeWave Design Environment](#)
- [PrimeWave Design Environment for Reliability Analysis](#)

Related User Documentation

For related documents, such as application notes, see the [PrimeWave Design Environment Documentation](#) page.

Downloadable Help

To download the PrimeWave Design Environment documentation on your local Windows computer, see the download instructions on [SolvNetPlus](#).

Release Notes and Installation

The PrimeWave Design Environment Release Notes provide information about new features, enhancements, changes that are introduced after the help was published. The release notes also provide information about known issues and resolved Synopsys Technical Action Requests (STARs).

The PrimeWave Design Environment Installation Notes provide instructions on how to install the PrimeWave Design Environment tool.

To view the release notes and installation instructions:

1. Go to [My Product Releases](#).
2. Locate and click **PrimeWave Design Environment**.
3. Click the release version of interest.

PrimeWave Design Environment

This section lists features and enhancements introduced in this release.

- [Enhancements to Design Parameterization and Analog Circuit Optimization Flow](#)
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- [Enhancements to Device Grouping](#)
- [Enhancement to Job Distribution](#)
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Enhancements to Design Parameterization and Analog Circuit Optimization Flow

The following are enhancements to design parameterization:

- Design Parameterization in Netlist Flow
- Analog Circuit Optimization in Parameterized Netlist
- Global Design Parameterization - Global design parameterization for multiple testbench in PrimeWave can be applied to all testbenches in the test suite if required. Parameterized variables are set as global in this mode and can be overridden in each single testbench. This streamlines the parameterization procedure and ensures consistent parameterization for multiple tabs in a test suite.

The following are the enhancements for the Analog Circuit Optimization Flow:

- An automated flow is introduced for huge corner setup and Monte Carlo variation, you can use worst cases to run optimization and reduce the runtime. This flow allows you to run multiple optimization in one automated flow. It automatically verifies and extracts the new worst cases for next run. For more information, see Automated Flow in Worst Cases.
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Enhancement to PrimeWave Regression

Beginning with W-2024.09, you can export regression run results as a CSV file from the Regression's Run History GUI.

For more information, see PrimeWave Regression.

Enhancements to Device Grouping

PrimeWave is enhanced to support device grouping in netlist based flow. You can select more than one instance, parameterize certain parameters and the parameterization is applied automatically to all selected instances.

For more information, see Design Parameterization in Netlist Flow.

Enhancement to Job Distribution

The job distribution is enhanced to support automated job regulation based on the available disk space. This feature provides enough time to you to clear the disk space without shutting down the entire CC-PW session. After, sufficient space is available, all the suspended simulations are resumed automatically and the progress is visible in Job Monitor.

For more information, see [Displaying Simulation Jobs](#).

The **Run post process with Simulation job** and **Rerun jobs upon unexpected exit** check boxes on the Job Definition and Job Management dialog box are enabled by default.

For more information, see [Job Monitor Tool](#).

Enhancement to Netlisting

Beginning with W-2024.09 release, you can use the shortcut icons on PrimeWave Design Environment window to display logs and netlists of the selected jobs. For more information, see [Viewing and editing netlists and Displaying the Simulation Log](#).

Capability to Change Data Driven Sweep Names

Beginning with W-2024.09 release, you can edit and rename Data Driven sweep names while setting up Parametric Analysis.

For more information, see [Setting Up Parametric Analyses](#).

Enhancements to Outputs View

Global output allows you to set Outputs that are common to all testbenches in a Multi-TestBench (MTB) session. You can set the selected outputs or expressions as local or global and add specifications for them.

For more information, see [Global Output and Specifications](#).

Enhancements to Sequential Testbench usage

When running the sequential testbenches with corners, if few corners fail in the 1st testbench, the dependent testbench does not have all the inputs needed for its simulation. You can rerun the failed testbenches using the Rerun option.

For more information, see [Sequential Testbenches](#).

Enhancements to Preferences

For more information about new and removed commands and command options, and changes to existing commands and command options in version W-2024.09 that might affect existing script, see *PrimeWave™ Design Environment Tcl Reference Manual*.

New Yield Function in Measurement Functions

A yield function that measures the count of the value specified by `spec_min` and `spec_max` and returns the value for $(\text{valid_count}/\text{total_count}) * 100$ is added.

For more information, see .

Enhancements to WaveView Preferences

All WaveView preferences are now recognized when running the commands, `db::getPrefValue` and `db::setPrefValue`.

For more information, see *PrimeWave™ Design Environment Tcl Reference Manual*.

New Parametric Function to Waveform Functions

A parametric function that gets the parametric plots of `measure` versus sweeping `param_name` is added.

For more information, see .

Library-Based Connect Rules

Beginning with W-2024.09 release, PrimeWave has an additional option to specify external design files in addition to library files. For more information, see *Setting Mixed-Signal Simulator Options*

Enhancements to Simulation Tools

Beginning with W-2024.09 release, you can create DC analysis in VCS PrimeSim AMS. For more information, see *Setting Mixed-Signal Simulator Options*.

You can now set the digital file options using **Digital File Options** window. For more information, see *VCS PrimeSim AMS DC Analysis*.

Refined Error and Warning Messages

The documentation of error and warning messages is refined to include detailed information about the messages.

For more information, see [PrimeWave Design Environment Error Messages](#).

Enhancements to Runtime Statistics of Monte Carlo Simulation

The Monte Carlo simulation with a single core or the **Multiple Runs** option, now displays the runtime statistics in the report.

For more information, see [Working With Monte Carlo Data Mining Results](#).

PAM-N Measurements

Beginning W-2024.09, PrimeWave Design Environment now includes PAM-N measurements. The following measurement functions are added to enable this enhancement:

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Recovering Simulation Progress

PrimeWave now allows you to exit your session and reconnect to simulation and measurement evaluation tasks while maintaining their status even after exit. This status is maintained upon voluntary exit as well as in the rare event of an unexpected shut-down.

For more information, see [.](#)

Independent Process to Manage Simulation and Post Process

While viewing the job processes that are running, the Job Monitor view options now display the status of jobs Running, Done, and Total (% of Done) while the simulation is running for different sections

For more information, see [.](#)

PrimeWave Design Environment for Reliability Analysis

The PrimeWave Reliability Environment is now a part of the PrimeWave Design Environment. For the W-2024.09 release, both the `primewave` and `primewave_reliability` executables are still available. Consult your Synopsys representative if you have any questions.

PrimeWave Design Environment for Reliability Analysis provides the following enhancements and changes on reliability applications in this release:

- [PrimeSim EMIR Application: New Features and Enhancements](#)
- [PrimeSim CCK Application: New Features and Enhancements](#)
- [PrimeSim Design Insights Application: New Features and Enhancements](#)
- [PrimeSim Custom Fault Application: New Features and Enhancements](#)

See also What's New in This Release for more information on PrimeSim Reliability enhancements or changes on the batch mode.

PrimeSim Design Insights Application: New Features and Enhancements

The PrimeSim Design Insights (DI) application adds the following enhancements:

- DI analysis automatic mode enhancement - identify pull-up or pull-down paths by sorting the P or N type drain or source terms together and all the gate terms together. In addition, the tool also identifies the paths such as InstPin to port, port to gate, and so on.

For more information, see [Advanced Design Exploration With Design Insights](#)

PrimeSim EMIR Application: New Features and Enhancements

The new PrimeSim EMIR GUI provides menu commands for setting up and running Library EMIR analysis, Dynamic EMIR analysis, Fast IR analysis, Static IR analysis, and static resistance check (SPRES), browsing analysis results, and invoking what-if analysis and least resistance path checking tools. The look and feel of the PrimeSim EMIR GUI is consistent with the look and feel of the Custom Compiler GUI.

For more information, see [PrimeSim EMIR Application: New Features and Enhancements](#)

PrimeSim CCK Application: New Features and Enhancements

The PrimeSim CCK application does not have any new feature in this release.

PrimeSim Custom Fault Application: New Features and Enhancements

The PrimeSim Custom Fault application does not have any new feature in this release.